Demian Kong

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Growing Agent Platforms · Verified Tool Use · Scientific AI

RESEARCH PROFILE

I build systems that turn human needs into executable, reusable capabilities. My work connects growing agent platforms, configurable scientific database agents, and verified Text-to-SQL. I am seeking Ph.D. opportunities in AI agents and scientific AI.

EDUCATION

University of Chinese Academy of Sciences / CNCB M.S. Bioinformatics · Sep 2023–Jun 2026

Thesis: *Research and Development of an Agent Framework for Bioinformatics Data Retrieval Based on Large Language Models.*

Jilin University B.S. Pharmacy · Sep 2019–Jun 2023

Thesis research on MKP5 in hepatic fibrosis, combining experimental models and bioinformatic analysis.

SELECTED RESEARCH & SYSTEMS

Chrona · Creator & Sole Developer 2026–Present · Source

- Building a local-first platform where agents create and revise persistent modules in response to human needs. Modules combine event-driven execution, persistent data, scheduled work, and routed tool calls.
- Implemented iterative build sessions, content-addressed versions, tests, platform-owned execution probes, and migration checks on data copies, with activation bound to reviewed versions.

Dingent · Founder & Lead Developer · NGDC, CNCB 2025–Jun 2026 · Source

- Designed a framework integrating eight CNCB biological data resources through MCP plugins; compiled visual workflows into checkpointed LangGraph state graphs, preserving entities and query scope across databases.
- First-listed co-first author of the bioRxiv preprint; technical support after graduation.

EWAS AI Assistant · Technical Lead & Developer · NGDC, CNCB 2024–2025

- Built entity-enhanced schema linking, AST-level SQL validation, and execution-guided query repair for the natural-language AI assistant of EWAS Open Platform.
- Designed a 30-question expert-reviewed benchmark with manually verified reference SQL over a fixed system version and database snapshot (project/thesis evaluation).

ProLLaVA · Independent protein–language prototype Oct 2024–Feb 2025

- Connected ESM2 to Llama-3-8B on UniProt sequence–function data; ran projector pretraining, 4-bit LoRA fine-tuning, and protein–text contrastive training with advisor-provided computing resources.

iDog & NGDC Data Applications · Co-first Author & Developer 2023–Jun 2026

- Performed canine transcriptome and DNA-methylation analysis; built DogRAG, GSA submission-data retrieval, and the Smart consultation plugin; contributed to BioKA biomarker curation.

SELECTED PUBLICATIONS

Kong D†, Bei S, Wu Y, Tang B†, Zhao W. Dingent: An Easily Deployable Database Retrieval and Integration Agent Framework. *bioRxiv preprint*, 2026. DOI

Yang F†, Xiong Z†, Zong W†, **Kong D†**, et al. EWAS Open Platform 2026. *Nucleic Acids Research*, 2026;54(D1):D1061–D1068. DOI

Liu Y†, Wang Y†, Sun J†, **Kong D†**, Zhou B†, et al. iDog: A Multi-Omics Resource for Canids Study. *Nucleic Acids Research*, 2025;53(D1):D1039–D1046. DOI

† Equal contribution / co-first author. Additional publications: BioKA (*NAR*, 2024) and NGDC database-resource consortium papers (2024–2026).

TECHNICAL PREPARATION & RECOGNITION

Systems: Python, TypeScript, SQL, LangGraph, MCP, FastAPI, React / Next.js, Docker, Nix, Linux. **ML:** PyTorch, Hugging Face Transformers, LoRA / QLoRA, contrastive training.

UCAS Graduate Academic Scholarship (2023, 2024, 2025). Project lead, Provincial Undergraduate Innovation Training Program, Jilin University (completed 2022).